

CardioCare Final Project Report

Machine Learning Final Project - Heart Disease Risk Prediction

ETHICS STATEMENT (required): This system only assists cardiologists and does not make autonomous diagnoses. All final medical decisions must be made by licensed physicians. Models may reflect dataset bias and must not replace professional medical judgment.

1. Problem Definition and Purpose

CardioCare predicts heart-disease risk from clinical features to inform, not replace, physician judgment (inform, not decide). In clinical practice, false negatives are more dangerous than false positives because a missed heart disease case can delay life-saving treatment.

2. EDA Key Results

The Cleveland subset (303 patients) shows moderate class imbalance, missing tokens encoded as '?', and outliers in chol, trestbps, and oldpeak. Target distribution motivated recall-focused evaluation instead of accuracy alone.

3. Preprocessing Decisions

Based on EDA: replace '?' with np.nan; use pd.isna() for detection; impute numeric features with median and categorical features with mode; drop columns only when missing rate exceeds 30%; clip outliers with train-only IQR bounds. train_test_split occurs before any fit operation to prevent data leakage.

4. Model Comparison and Final Selection

Three families were compared with MLflow: Logistic Regression, SVC, and Random Forest. 5-fold CV and grid search optimized recall. Final hold-out metrics: recall=0.929, precision=0.812, balanced_accuracy=0.873, f1=0.867. Logistic Regression was selected because it achieved the highest recall (0.929), minimizing missed disease cases.

5. Testing and Packaging

Unit tests verify: (1) prediction shape, (2) probability range and row sums, (3) clinical input validation, and (4) deterministic output under seed=42. Dockerfile uses python:3.10-slim and serves batch inference via sample_batch.csv.

6. Drift Detection and Retraining Plan

KS tests flagged features: ['chol', 'oldpeak']. Balanced accuracy dropped from 0.873 to 0.843. Retraining policy: trigger when KS p-value < 0.05 AND recall drops > 5%, with cardiologist review before deployment to prevent runaway feedback loops.

7. Serving Architecture

Model-as-a-Service (MaaS) is preferred: centralized PHI control, easier model updates, and acceptable latency for hospital batch scoring. On-device deployment would improve privacy but complicate model refresh.

8. Limitations, Ethics, and Future Work

Limitations include small sample size, single-center bias, and no prospective validation. With one additional

week: SHAP explainability, shadow-mode deployment, and external dataset validation would be added.

Model Comparison Table

logistic_regression: recall=0.929, precision=0.812, f1=0.867, balanced_accuracy=0.873

svc: recall=0.893, precision=0.758, f1=0.820, balanced_accuracy=0.825

random_forest: recall=0.893, precision=0.833, f1=0.862, balanced_accuracy=0.871

Core Figures (EDA / Evaluation / Drift)

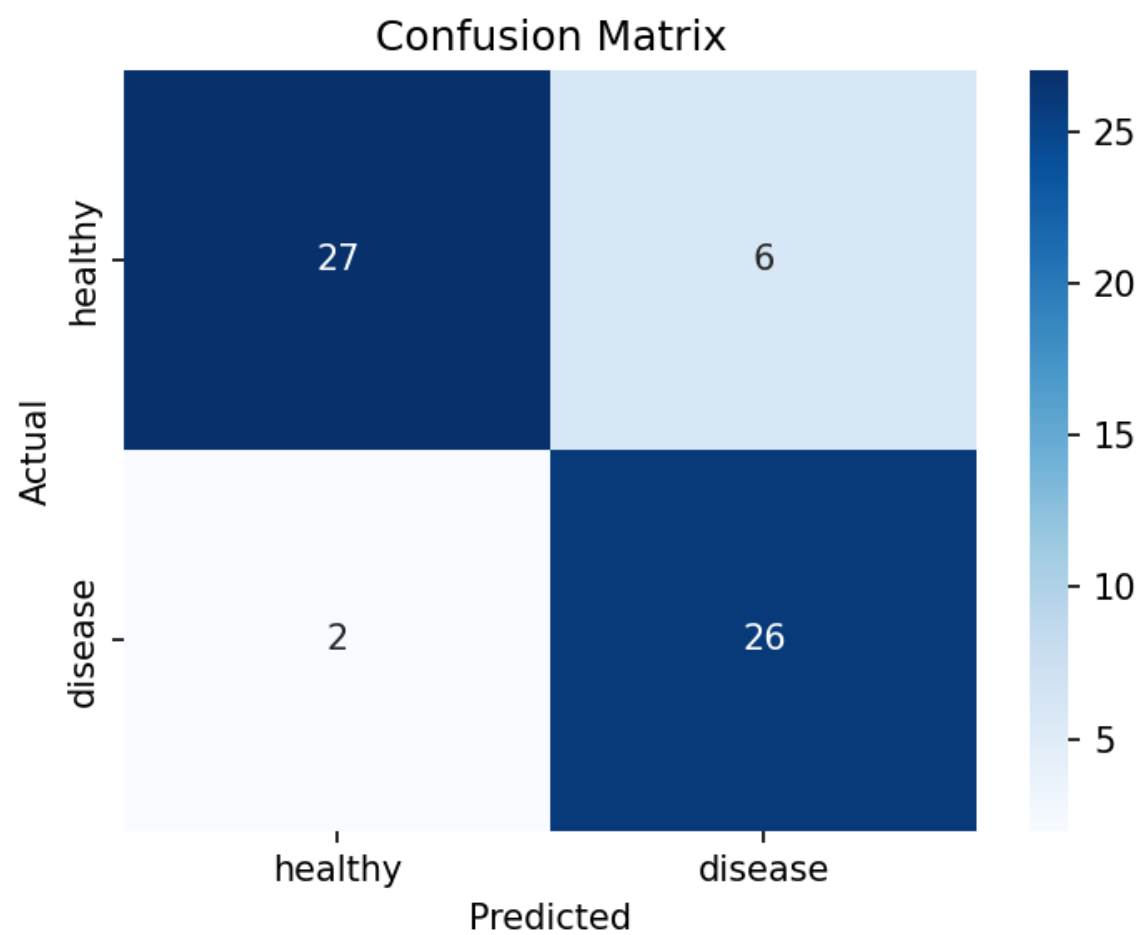


Figure 1. Confusion matrix on the held-out test set.

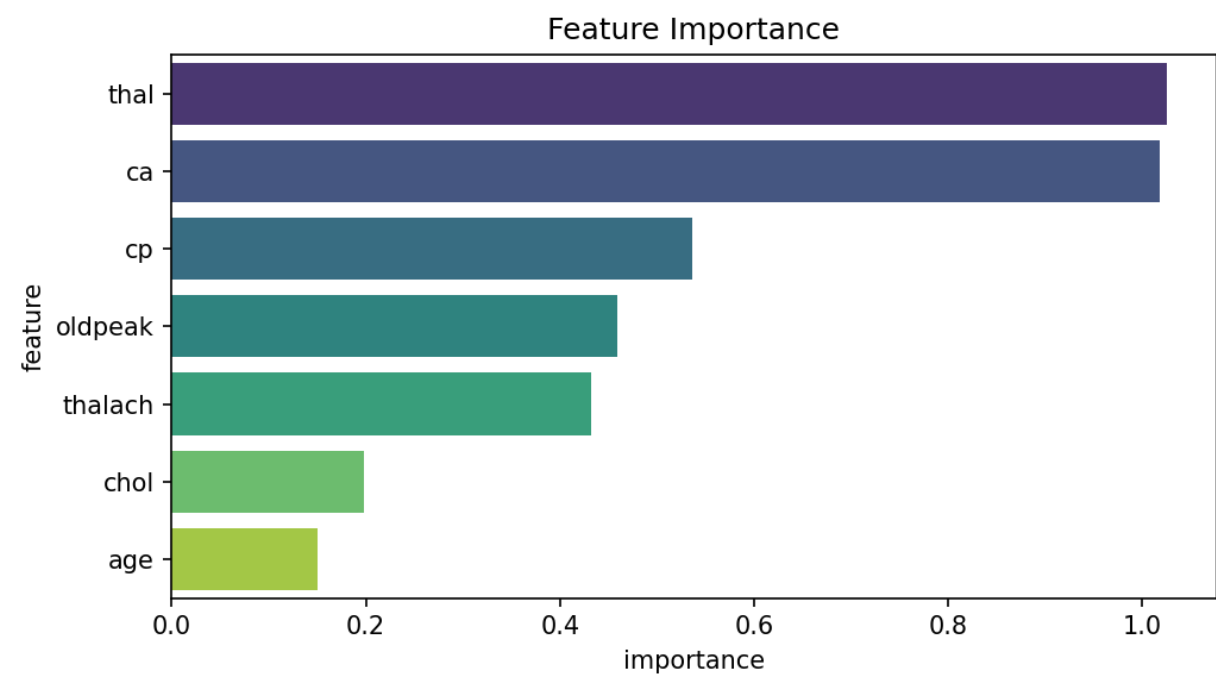


Figure 2. Feature importance. Top features: thal (1.03), ca (1.02), cp (0.54), oldpeak (0.46), thalach (0.43).

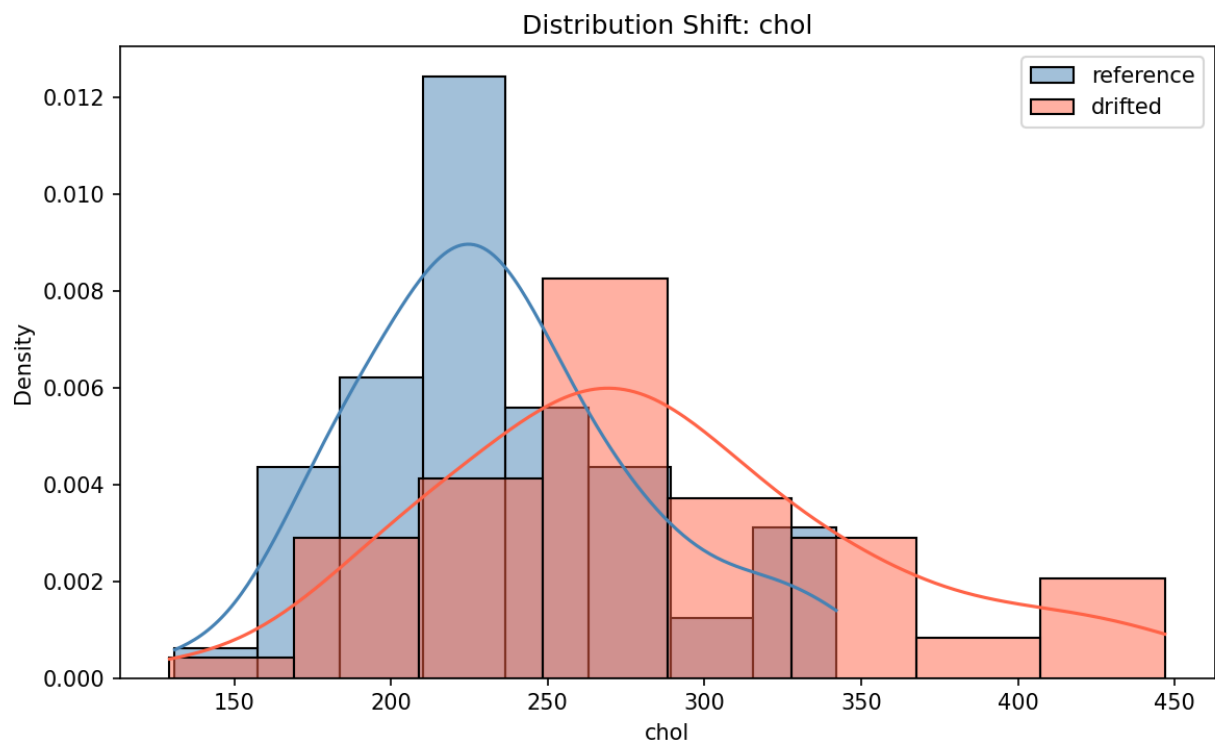


Figure 3. Distribution shift for chol (reference vs drifted test set).

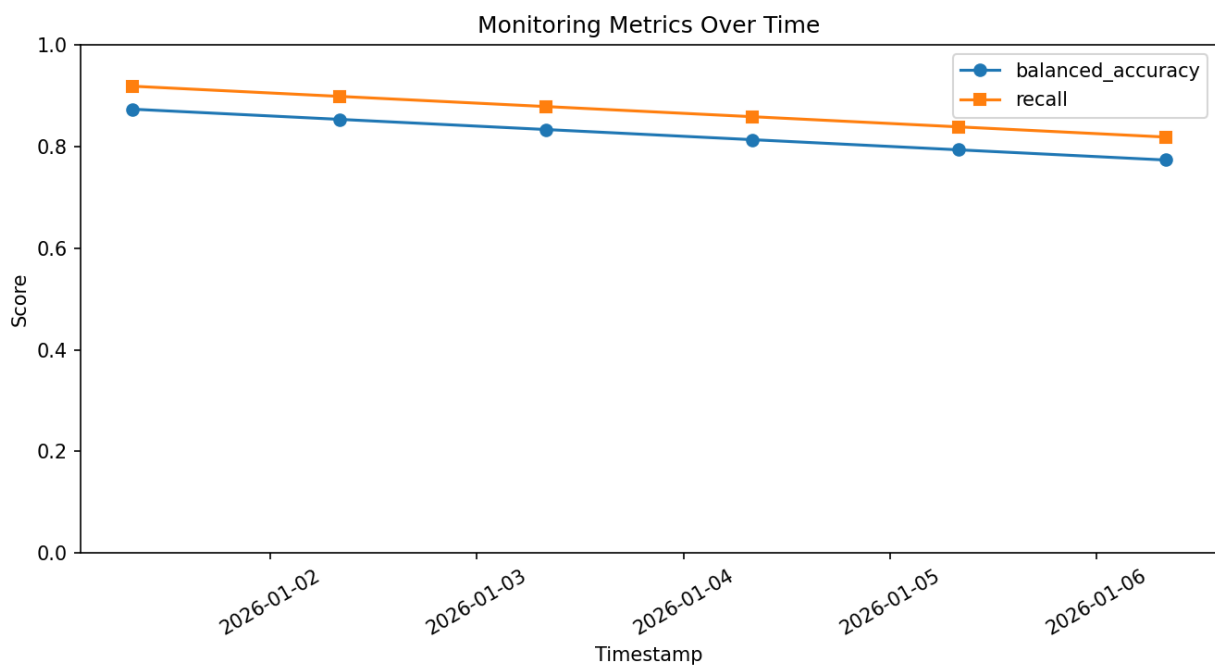


Figure 4. Synthetic monitoring time series for balanced accuracy and recall.

Bonus: DVC Data Versioning (+5)

This project can be extended with DVC: Git tracks code, while DVC tracks large datasets and model binaries. That separation solves Git's large-file limitation and enables full experiment replay.

Bonus: MLOps Closed Loop

CI/CD/CM/CT loop: Code commit -> CI unit tests -> CD Docker deploy -> CM drift monitoring (KS + recall) -> CT retrain trigger -> human clinical review -> model registry update.

Feature Store / Model Registry (Conceptual)

Feature Store candidate: chol (drift-prone, clinically meaningful). Model Registry metadata: training_data_hash, recall_on_holdout, random_seed, selected_features, and approval_status.

Appendix: AI Tool Usage Disclosure

This project used AI coding assistants (Cursor/ChatGPT) for boilerplate generation and debugging. All core logic, experiments, metrics, and design decisions were verified by the author. Copied code snippets include source links in code comments.

ETHICS (closing): This system only assists cardiologists and cannot make final medical decisions independently. All predictions require physician review.